Excess entropy and divalent-ion mobility in multicomponent charged LJ mixtures

HNC screening · replicated MD · thermodynamic integration, pair entropy and 2PT

+0.30 kB

excess entropy gained by HNC-selected solvent designs

+18–20%

divalent-ion self-diffusion, both signs

0%

effect of adding labels without changing interactions

Design: label count vs interaction diversity

1

HNC screen

35 integer-charge, composition and dispersion candidates at $T^* = 2$, $\rho^* = 0.7$; 8 selected

2

Replicated MD

8 mixtures × 3 matched replicas, N = 4000, x(±2) = 0.03; probes matched across mixtures

3

Entropy and transport

TI over 19 densities (fixed LJ cutoff, Debye–Hückel endpoint); MSD in one common COM frame

Code	Mixture (all with 3% of each ±2 probe)	Role
B3	94% LJ solvent	Baseline
B5	B3 + 5% of each ±1 ion	Salt addition
U4	64% solvent + 30% weak solvent, ε = 0.25	Reference
L5	U4 with 2 identical weak-solvent labels	Label null
M5	U4 split into ε = 0.0625 / 0.5625	Moderate split
H5	U4 split into ε = 0.01 / 0.81	Strong split
U6 / H7	U4 / H5 plus 5% of each ±1 ion	With salt

U4 → M5 → H5 keep charged composition, ionic strength and the mean bare LJ amplitude $\Sigma_{ij} x_i x_j \varepsilon_{ij} = 0.7225$ fixed

Seven kinds of particles, one size

Solvent A
q = 0, ε = 1

all mixtures, 54–94%

Weak solvent B
q = 0, ε = 0.25

U4, L5, U6: 30%

Split solvent B_1 / B_2
q = 0, ε = 0.01 / 0.81 (H)
or 0.0625 / 0.5625 (M)

M5, H5, H7: 15% each

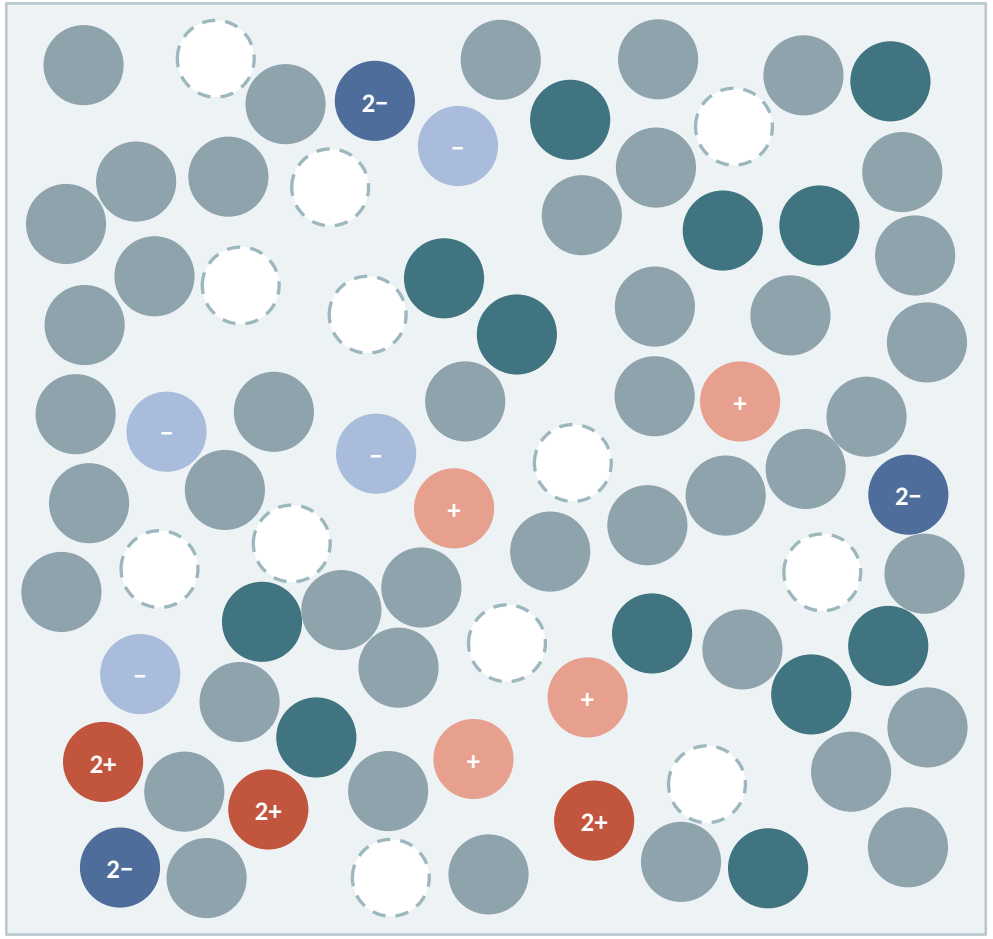
Divalent probe ions
q = ±2, ε = 1

all mixtures: 3% each

Monovalent background
q = ±1, ε = 1

B5, U6, H7: 5% each

All particles: σ = 1, m = 1; $\varepsilon_{ij} = \sqrt{(\varepsilon_i \varepsilon_j)}$; Coulomb with Bjerrum length 0.5σ at $T^* = 2$



Cartoon of H7 (2D sketch with roughly its proportions; not simulation data)

HNC for charged mixtures in four steps

1 **Pair structure** $g_{ij}(r),\ h_{ij} = g_{ij} - 1$

Probability of finding a j particle at distance r from an i particle, relative to a random mixture

2 **Ornstein–Zernike equation** $h_{ij} = c_{ij} + \Sigma_k \rho_k c_{ik} \otimes h_{kj}$

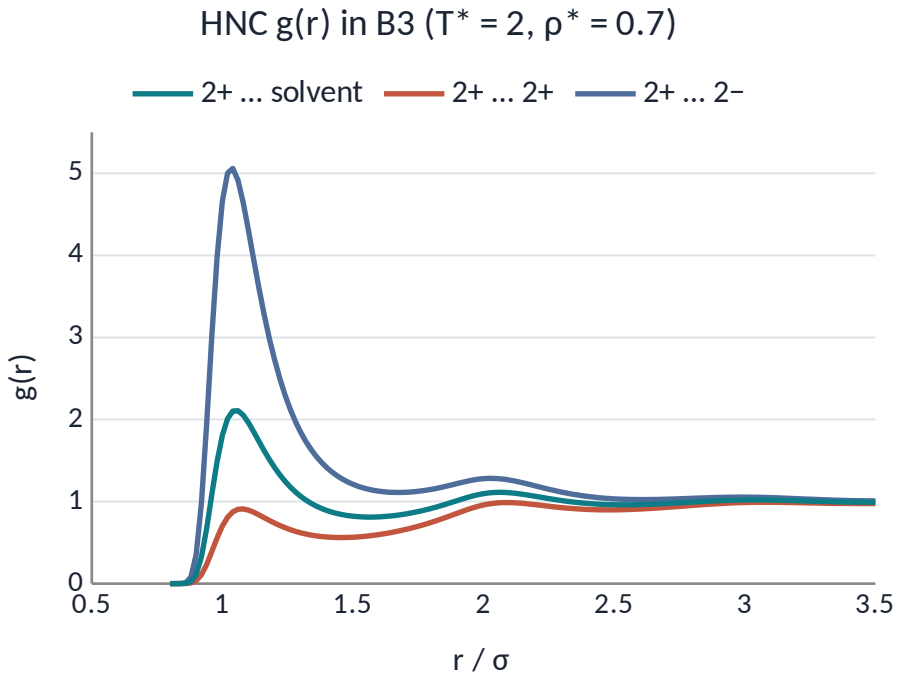
Total correlation = direct part + indirect part relayed through every species k (a p × p matrix equation in k space)

3 **HNC closure (Morita–Hiroike)** $g_{ij} = \exp(-\beta u_{ij} + h_{ij} - c_{ij})$

Second relation that closes the system; it neglects the bridge diagrams and is accurate for ions

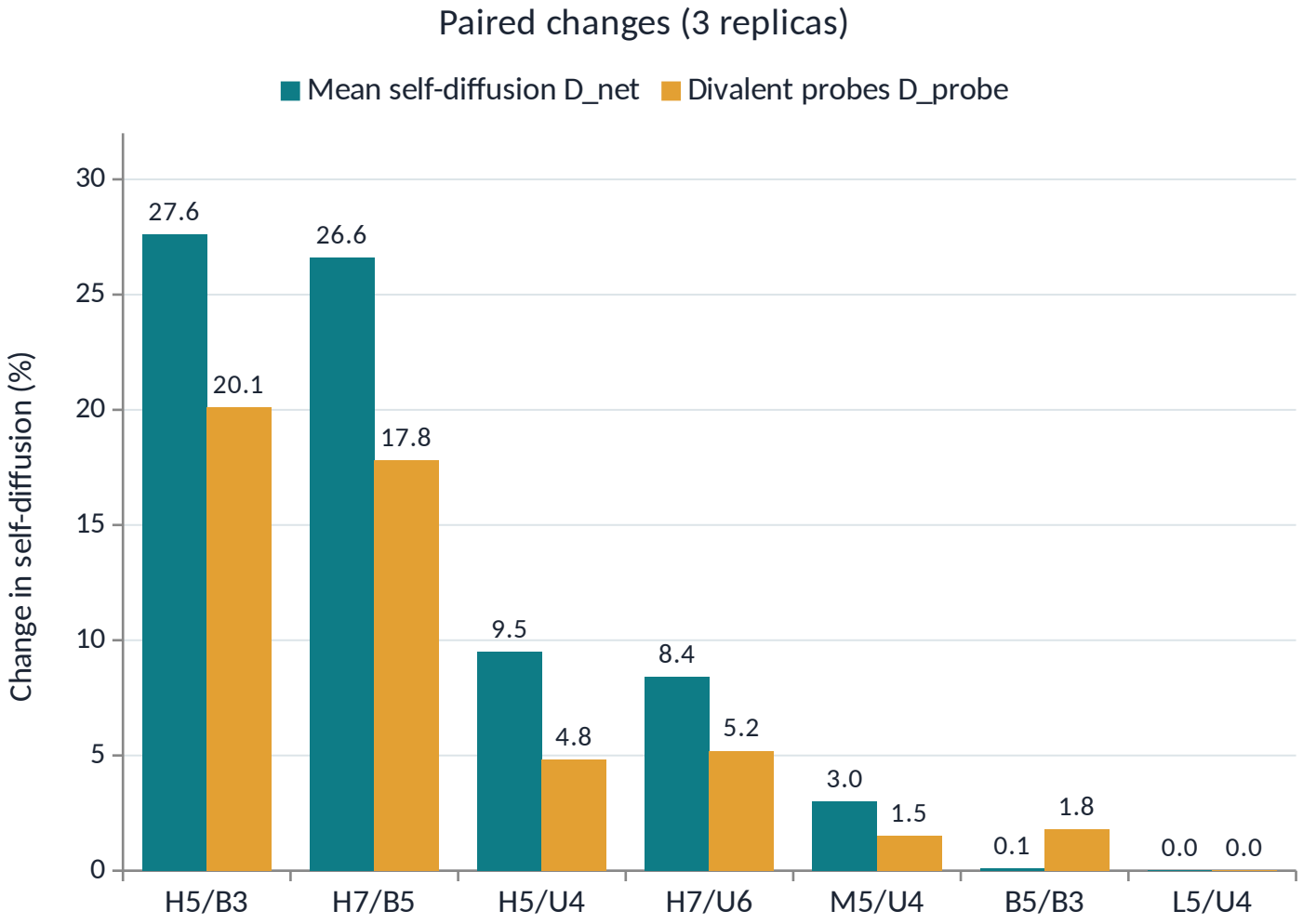
4 **Charges (Allnatt; Lomba–Høye)** $c_{ij} \to -\lambda_B q_i q_j / r$

Coulomb tails decay too slowly to Fourier transform: subtract the screened (Debye) tail analytically and solve for short-ranged parts



Why it is useful here: in HNC the free energy follows in closed form from g and c, so $s_{ex} = u_{ex} - a_{ex}$ costs one solution per mixture, enough to screen dozens of compositions

Entropy speeds up ions, less than the solvent



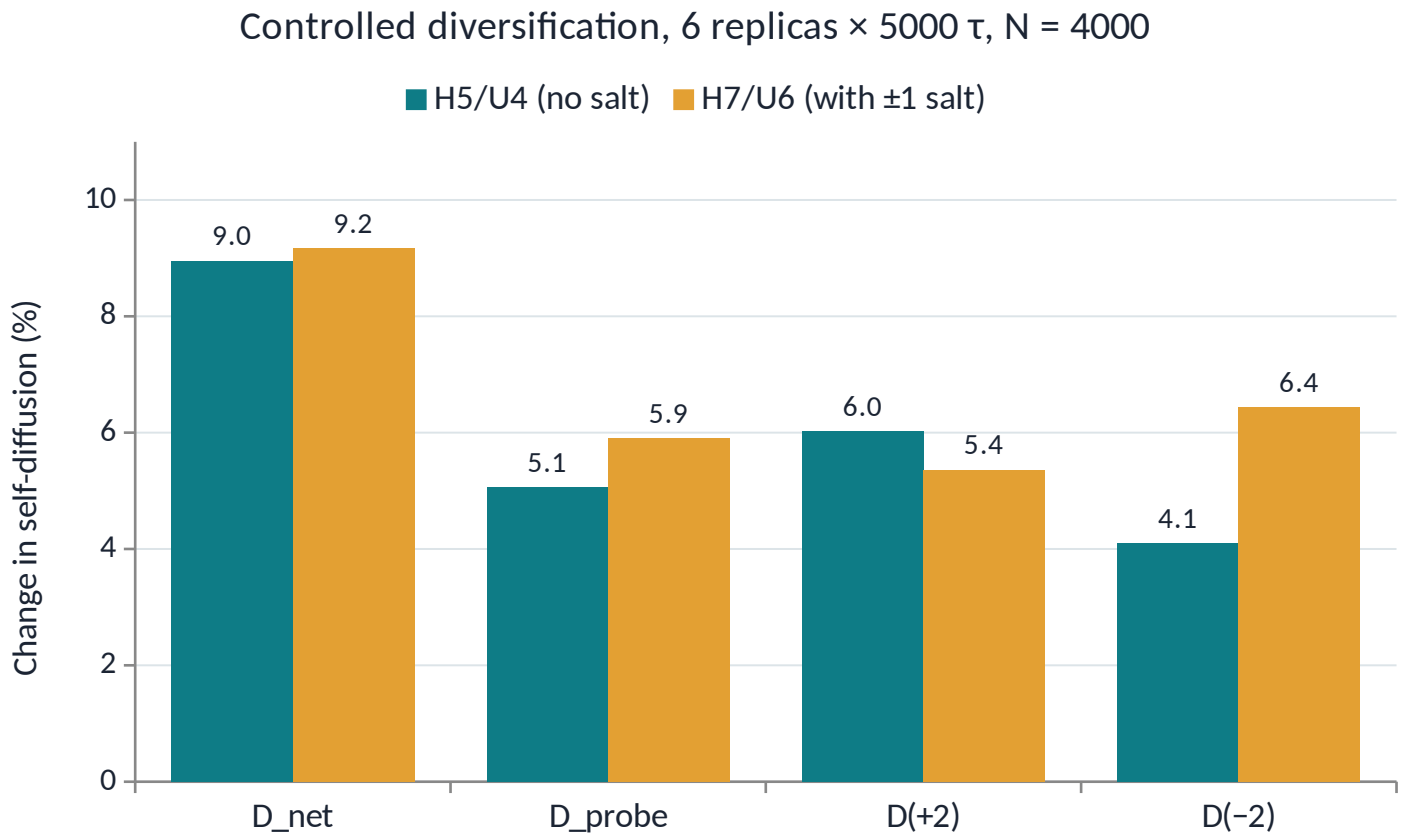
Full designs: Δs_ex ≈ +0.30 kB, probes +18–20%; each sign accelerates separately

Controlled diversity: Δs_ex = +0.103 kB, mean +8–9%, probes +5–6% (resolved by longer runs, next slide)

Labels only (L5/U4): identical diffusion in every replica

Salt only (B5/B3): Δs_ex slightly negative, no resolved speedup

Longer runs resolve a +5–6% ion response



Probe 95% CIs: H5/U4 [+4.0, +6.1]%, H7/U6 [+4.7, +7.1]%. H7/U6 is stable in every MSD window; H5/U4 drops to ≈ +3% beyond 100 τ, so it is quoted as +3 to +5%.

−7%

shear viscosity η (Green–Kubo), U4 → H5 and U6 → H7

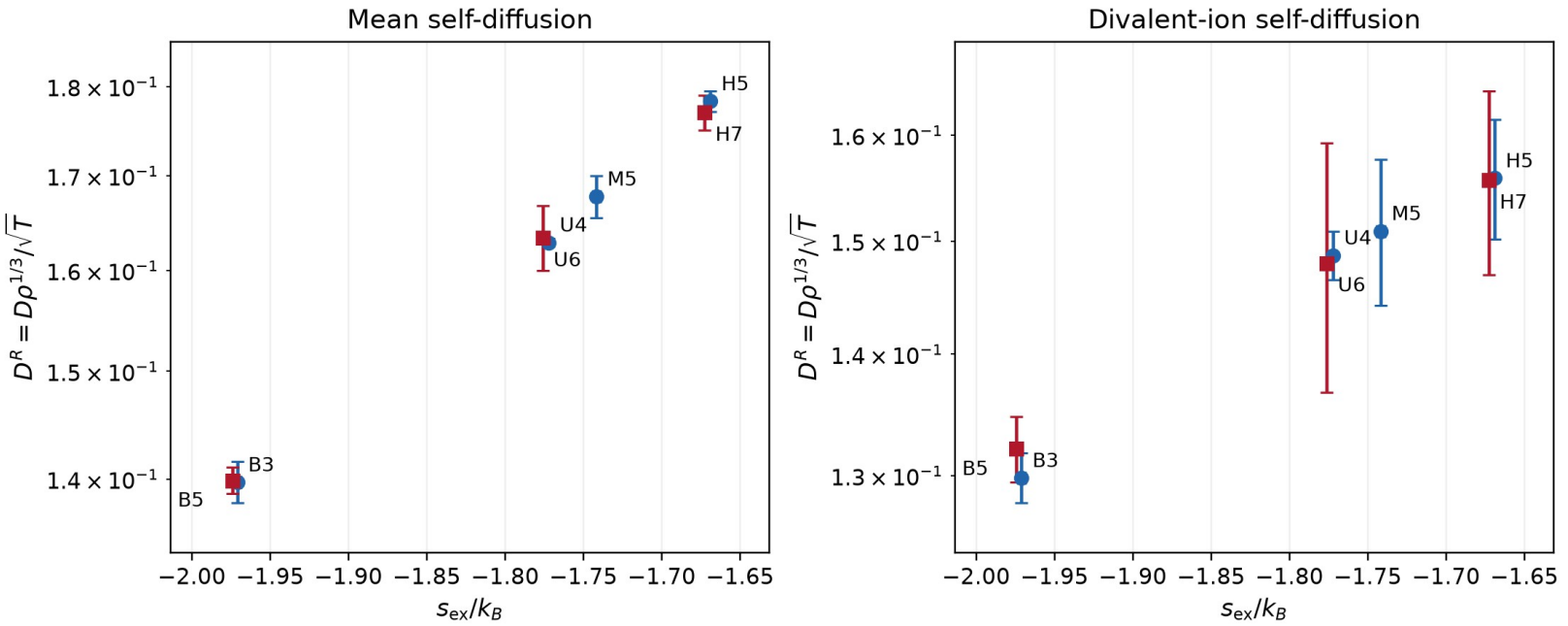
+1% vs −2 to −3%

D_net·η stays put (Stokes–Einstein); D_probe·η falls: probes only partly carried along

+5.3% / +6.0%

same probe contrasts at N = 8000; D shifts +0.8–1.3%, as Yeh–Hummer predicts

TI entropy tracks mobility; HNC and s2 follow



Descriptive Rosenfeld fits (3–4 mixtures, $R^2 > 0.99$): B_net ≈ 0.8, B_probe ≈ 0.6. Controlled local probe slopes ≈ 0.6 × the mean-fluid value. Absolute HNC entropies are 0.06–0.09 kB too low; increments are much better.

≤ 7%

HNC error on the controlled entropy increments

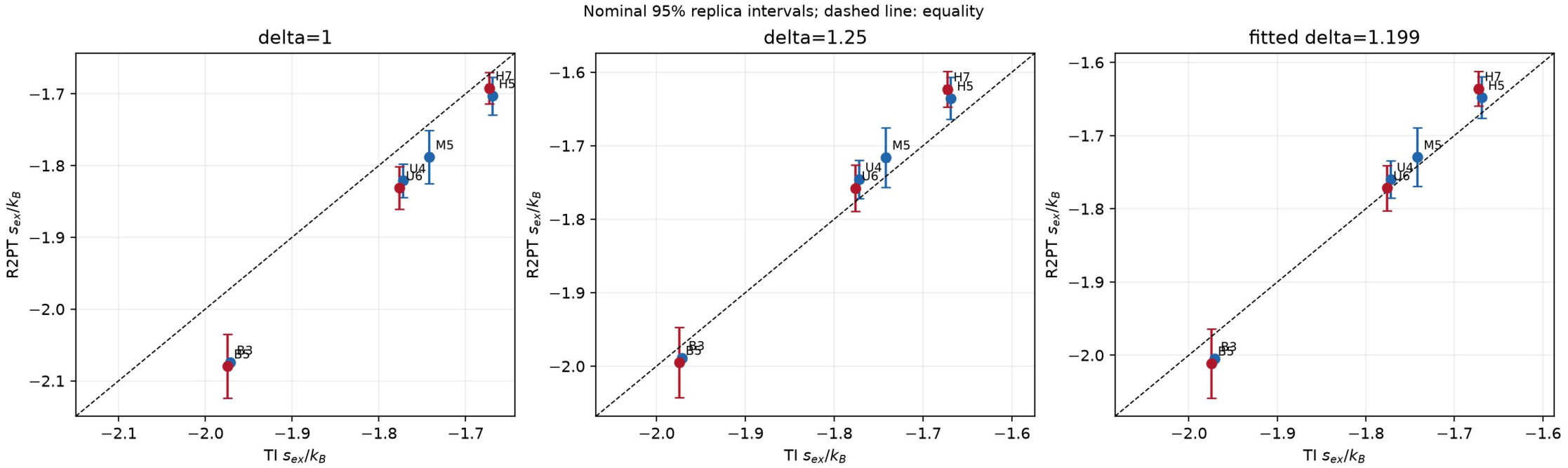
≈ 80%

share of s_ex carried by the MD pair entropy s_2, and of its increments

−0.33 kB

beyond-pair residual s_ex − s_2: composition dependent, not negligible

2PT when TI is unfeasible: calibrate, then rank



Calibrate one δ

Fit over 7 mixtures: $\delta^* = 1.20$ [1.17, 1.23]; $\delta = 1.25$ cuts RMS bias 0.067 → 0.029 kB; held-out bias 0.030 kB

Good within a family

M5/U4 and H5/U4 increments reproduced within 0.01 kB; label split exactly zero

Limits

H7/U6 +30%, baseline → design +17–23%; D enters the fluidicity, so 2PT cannot test an entropy–diffusion link on its own

Next: fast entropy estimators for electrolytes

2PT, velocity based

Ready

- Calibrated δ ≈ 1.2–1.25 for this model family
- One dense trajectory per candidate
- Use for ranking; recalibrate on TI anchors
- Not independent of diffusion

Voronoi cluster entropy

Open

- 24 trajectories packaged with TI, s_2 and D references
- Must reproduce TI order, the 0.03 and 0.10 kB increments and the label null
- Structure based, so independent of transport

1. **HNC screen**
   candidate mixtures
2. **TI anchors**
   2–3 states to calibrate and validate
3. **2PT / Voronoi**
   rank many candidates
4. **Conductivity**
   Green–Kubo current, not only self-diffusion

Next physics test: probe–solvent coordination, association lifetimes and exchange kinetics — does weaker local caging control the ion mobility?